

AMANDA ASH

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EDUCATION

- Ohio State University, Columbus, OH** Fall 2021 - Current
PhD Candidate in Astronomy
- Ohio State University, Columbus, OH** Fall 2021 - Fall 2024
Masters in Astronomy
- University of North Georgia, Dahlonega, GA** Spring 2018 - Spring 2021
Bachelor of Science
Major in physics, minor in mathematics

RESEARCH EXPERIENCE

- Rotation and Magnetism in Low Mass Stars** Spring 2024 - Current
Ohio State University
Advisors: Marc Pinsonneault, Lucy Lu
- Test theoretical models of stellar spin down against observational data including starspot measurements and gyro-kinematic ages.
 - Funded through NASA ADP
- Seismology of Luminous Giants** Fall 2021 - Fall 2024
Ohio State University
Advisors: Marc Pinsonneault, Mathieu Vrad, Joel Zinn
- Used stellar evolution models to determine the source of discrepancy between stellar radii determined from seismology and *Gaia* radii for luminous giants
 - Used Modules for Experiments in Stellar Astrophysics (MESA) to generate model grids and compare to APOKASC data
- Helium Enrichment in Globular Cluster Multiple Populations** Summer 2020
University of North Georgia, collaboration with Dartmouth College
Advisor: Gregory Feiden
- Investigated the effect of varying Helium abundance in the thermal structure of stellar model atmospheres
 - Used the MARCS model atmosphere code to generate model grids
- Magnetic Activity Variability in Praesepe Open Cluster** Summer 2019
Center for Astrophysics | Harvard & Smithsonian, NSF REU Program
Advisor: Stephanie T. Douglas
- National Science Foundation funded Research Experience for Undergraduates program
 - Used statistical analysis techniques to determine whether magnetic variability in low mass stars is driven by rotation.
- Impact of Starspots on Observable Stellar Properties** Summer 2018 - Spring 2021
University of North Georgia, Physics and Astronomy Department
Advisor: Gregory Feiden
- Compared predictions from a flexible starspot model imposed on the Dartmouth Stellar Evolution Model to Gaia data for open clusters

- Used statistical techniques, including ensemble sampling, to measure starspot properties for open clusters.
- Research funded through an institutional grant

FIRST AUTHOR PUBLICATIONS

Calibrating Stellar Spin Down in Old Partially Convective Stars **Ash, A.**, Pinsonneault, M., Lu, L. (2025) (Manuscript in Preparation)

Testing the Breakdown of the Asteroseismic Scaling Relations in Luminous Red Giants **Ash, A.**, Pinsonneault, M., Vrad, M., Zinn, J. (2025) ApJ, 979, 135

ADDITIONAL PUBLICATIONS

The YREC Stellar Evolution Code: Public Data Release. Marc H. Pinsonneault, Jennifer L. van Saders, Lyra Cao, Jamie Tayar, Franck Delahaye, Leslie M. Morales, Rachel A. Patton, Matthew C. Rendina, Joel C. Zinn, Zachary R. Claytor, **A. Ash**, Michael Byrom, Kaili Cao, Vincent A. Smedile

Anchoring Stellar Age Indicators: A Cross Calibration of [C/N] and Gyrochronology Ages via the AgeVelocity Dispersion Relation. Lu, Y., Pinsonneault, M., Ting, Y., Van-Lane, P., Roberts, J. , Tayar, J., Stone-Martinez, A., **Ash, A.** (2026) ApJ, 171, 1

Chemically Self-Consistent Modeling of the Globular Cluster NGC 2808 and its Effects on the Inferred Helium abundance of Multiple Stellar Populations. Boudreaux, E. M., Chaboyer, B. C., **Ash, A.**, Hoh, R. E., and Feiden G. (2024) ApJ, 980, 180

CONTRIBUTED TALKS

Ash, A., Pinsonneault, M., Lu, L. . (2025, June). *Rigorous Tests of Stellar Spin Down in Low Mass Stars* Presentation at the 2025 SDSS Collaboration Meeting | Max Planck Institute for Astronomy, Heidelberg, Germany.

Ash, A., Douglas, S. T. (2019, August). *Rotationally Modulated Magnetic Variability in Praesepe K and M Dwarfs*. Presentation at the 2019 Center for Astrophysics | Harvard & Smithsonian NSF REU Symposium, Center for Astrophysics | Harvard & Smithsonian.

Ash, A., Hamilton, J., & Feiden, G. (2019, March). *A Flexible Model for Investigating the Properties of Starspots: Comparison of Models to Observed Data*. Presentation at the meeting of the Southern Atlantic Coast Section of the American Association of Physics Teachers, University of North Georgia.

POSTER PRESENTATIONS

Ash, A., Pinsonneault, M., Vrad, M., Zinn, J., (2024). *Testing the Breakdown of the Asteroseismic Scaling Relations in Luminous Red Giants* Poster presented at The SDSS Collaboration Meeting, Las Cruces, NM.

Ash, A., Douglas, S. T., Núñez, A., Morris B., Agüeros, M., Saar, S., (2019). *Rotationally Modulated Magnetic Variability in Praesepe K and M Dwarfs*. Poster presented at The 235th Winter Meeting of the American Astronomical Society, Honolulu, HI.

Ash, A., Douglas, S. T., Núñez, A., Morris B., Agüeros, M., (2019). *Rotationally Modulated Magnetic Variability in Praesepe K and M Dwarfs*. Poster presented at The Georgia Regional Astronomy Meeting, Dalton State College.

Ash, A., Hamilton, J., & Feiden, G. (2018). *A Flexible Model for Investigating the Properties of Starspots: Comparison of Models to Observed Data*. Poster presented at The National Conference for Undergraduate Research, Kennesaw State University.

Ash, A., Hamilton, J., & Feiden, G. (2018). *A Flexible Model for Investigating the Properties of Starspots*. Poster presented at The Georgia Undergraduate Research Coalition 1st Annual Posters at the State Capitol, Georgia State Capitol.

Ash, A., Hamilton, J., & Feiden, G. (2018). *A Flexible Model for Investigating the Properties of Starspots*. Poster presented at The Georgia Undergraduate Research Conference, University of North Georgia.

Ash, A., Hamilton, J., & Feiden, G. (2018). *A Flexible Model for Investigating the Properties of Starspots*. Poster presented at The Georgia Regional Astronomy Meeting, Emory University.

Ash, A., Hamilton, J., & Feiden, G. (2018). *A Flexible Model for Investigating the Properties of Starspots*. Poster presented at The Southern Atlantic Coast Section of the American Association of Physics Teachers Meeting, Clemson University.

Ash, A., Hamilton, J., & Feiden, G. (2018). *A Flexible Model for Investigating the Properties of Starspots*. Poster presented at The 20th Cambridge Workshop on Cool Stars, Stellar Systems, and the Sun ("CS20"), Boston University.

AWARDS

2020 Barry Goldwater Scholar

Honorable Mention, Chambliss Undergraduate Achievement Award, AAS 235

2018-2020 UNG Physics Department NASA Space Grant Fellowship

HONORS

Board of Regents 2021 Academic Recognition Day Representative for UNG

University of North Georgia Siler Scholar

2021 University of North Georgia Physics Department Outstanding Graduate

PROJECTS

Simulations of M-dwarf Convection

Fall 2020 - Spring 2021

University of North Georgia, Senior Capstone Project

- Used Python and the open source Python software, *Dedalus*, to simulate M-dwarf convection in a Cartesian coordinate system
- Imposed the stellar structure from the Dartmouth Stellar Evolution model onto the simulation

TEACHING

Teaching Coursework

Summer 2025

The Ohio State University Department of Astronomy

- Took specialized course in teaching college astronomy
- Developed course plans, curriculum, and curated lessons in stellar astronomy
- Presented interactive curriculum to undergraduate astronomy students and assessed learning gains

Teaching Assistant

Fall 2022

The Ohio State University Department of Astronomy

- Assist students in major specific astronomy courses
- Conduct office hours and exam review sessions

Learning Assistant

Spring 2019 - Spring 2021

University of North Georgia, Physics and Astronomy Department

- Assisted students during physics, astronomy, and interdisciplinary lab courses
- Trained in pedagogical techniques
- Participated in lesson planning alongside professors

Lab Assistant

Fall 2017 - Spring 2018

Kennesaw State University, Physics Department

- Supervised students during physics I and II labs
- Maintained and organized the lab environment
- Repaired and cataloged broken lab equipment

OUTREACH AND SERVICE

Graduate Representative

Fall 2024 - Current

Elected Position

- Collect feedback from astronomy graduate students and serve as an advocate.
- Attend meetings with faculty to discuss ongoing graduate feedback and concerns.

OSU Planetarium

Fall 2021

Presenter

- Presented ≈ 100 planetarium shows to a wide range of audiences including K-12 groups and the general public
- Developed planetarium events and programming such as themed shows and classroom demonstrations

University of North Georgia Astronomy Club

Spring 2019 - Spring 2021

President

- Coordinate public education opportunities with the Dahlonega Science Festival
- Organize campus wide events, e.g. campus wide stargazing nights

Senior Research Ambassador

Fall 2019 - Spring 2021

- Distribute information via workshops and seminars to undergraduate students at the University of North Georgia
- Act as a student mentor in the University of North Georgia summer research program

Through their Lens: The Hidden Pioneers of Astronomy

Spring 2021

- Designed and narrated a virtual observing night centered on women's contributions to women in astronomy.
- Contributed to the Virtual Dahlonega Science Festival
- Collaboration between the University of North Georgia Astronomical Observatory and the University of North Georgia Women in Physics Chapter

University of North Georgia Women in Physics

Spring 2021

Member

- Combined science and art to develop a series of posters highlighting women's contributions to physics and astronomy.
- Raised awareness for inclusivity and diversity in the field of physics